Development of Immersive Pedestrian Simulator with Dynamic Virtual Traffic

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***Abstract*- Pedestrian safety is a generic issue with approximately 20 pedestrians being injured every hour in Europe alone. This study is the framework explaining how an interactive Virtual Reality (VR) pedestrian simulator was created to assess and influence pedestrian behavior for various traffic conditions. Created on Unity game engine and on integration with Meta Quest 3, the simulator simulates real-like street crossing situations of varying difficulty levels in such a way that the user gets instant feedback. The system employs scene management as well as an interactive gameplay feature to provide an immersive experience. This can be used additionally to study the behavior of pedestrians for training, safety purposes and to provide data for autonomous vehicle testing. The simulator features three progressively challenging levels of increasing complexity ranging from single-lane road crossings with moderate traffic volumes to multi-lane roads with variable traffic flow.**

***Keywords*- Pedestrian Simulator, Virtual Reality(VR), Simulation, Unity, VR Integration, Behavioural Analysis**

I. Introduction

Pedestrian safety is a critical concern at the crossroads between transportation engineering, behavioural science, and urban planning. Walking is the easiest and most widespread mode of mobility, but pedestrians are among the most vulnerable road users as they lack physical protection and are directly exposed to traffic. According to the Federal Statistical Office [1], in 2023, 437 pedestrians lost their lives in traffic accidents, representing 15.7% Fatalities occurred overall that year. Across Europe [2] the figures were high, with pedestrians alone accounting for 18% of traffic deaths that occurred in 2022.Globally it is further intensified, based on the data reported by WHO [3] in 2023 21% of the fatalities occurred were pedestrians.These findings highlight the urgent need to improve pedestrian safety.

A. Motivation

Pedestrian vulnerability has been a persistent problem with urban transportation networks because pedestrian fatalities account for a significant share of traffic fatalities worldwide. As road networks and traffic become even more complex, pedestrian vulnerability only grows further because pedestrians have no physical protection [4]. Adding to this problem continues to be conventional training, awareness, and education in road safety: these interventions commonly do not recreate in their subjects genuine in-road interaction and hazard scenarios. Recent breakthroughs in immersive technologies such as virtual reality (VR) have discovered new applications for mimicking traffic environments under controlled yet safe settings. Pedestrians' behavior can be examined under varying traffic conditions through simulations based on VR. These can also be used as training tools to enhance decision-making skills and situation awareness of the users. Static models or observational studies lack interactive dynamism, adaptive difficulty levels, and real-time feedbacks that can be achievable under simulated VR settings. These render VR settings appropriate for applications that are safety-critical. To address these issues, this study introduces a 3D pedestrian simulator in a virtual reality environment in which subjects with a head-mount display must cross intersections under progressively adverse conditions. The simulation consists of levels of increasing complexity—including variables of varying traffic velocity, car densities, and ambient distractions—that closely mirror in-real-life cross situations. Not only can this system provide a controlled setting for behavioral research but it can also support user training and testing in an immersive, repeatable, and scalable fashion. Using virtual reality as a testing site, this work aims to move forward in developing improved pedestrian safety behavior and smarter simulation tools that can be utilized in informing urban planning, pedestrian education for safety behavior, and potential autonomous vehicle interaction platforms.

B. Research Objective

The final objective of this work is to design and evaluate a pedestrian simulator that can simulate in a realistic manner road-crossing scenarios under various difficulty levels. The pedestrian simulator must create user awareness, allow for

behavior investigation, and serve as a starting point for eventual usage in transportation safety applications and intelligent transportation planning. The detailed research objectives are as follows: To develop a 3D virtual environment in which individuals can interactively conduct road-crossing behaviors with a head-tracking display (VR head-mounted display). To perform a variety of simulation levels with increasing levels of difficulty incorporaing variables such as traffic congestion, vehicle moving speeds, crossing widths, and distractions. To measure user performance, reaction time, and decision-making skills for diverse levels of complexity in the simulator. To investigate pedestrian behavior under simulated traffic scenarios for potential application in real-world safety education, risk modeling, and interaction tactics for AVs. To demonstrate application of immersive simulation for learning and training pedestrian safety in a method with strong reproducibility, scalability, and engagement.

II. Literature Review

The knowledge on how people cross roads is important for design of safer traffic systems and for modeling of realistic training or simulation environments. Virtual reality (VR) is particularly good at this, since it can help us simulate real-life scenarios without risking the real-world consequences, but instead in a safe and contained environment—ideal for research and interactive learning. Doric et al. [4] designed a virtual reality simulator to study how risk was assed by individuals and make decisions in crossing the road as they are approaching traffic. The users are provided with a suit and an Oculus Rift headset to provide participants can have a realistic first-person experience. The research experimented with various scenarios with varying spacing between vehicles and found that individuals change their behavior in ways that demonstrated fatalities—particularly when cars were approaching fastly. This help them to demonstrates how fascinating simulations can help us better learn human reactions to high-pressure situations. Building on this, Holdgrün et al. [5] used full-body motion tracking and collision detection in a VR setup where the users crossed roads under different traffic conditions. If a virtual collision occurred, their posture at impact was noted and calculated using tools such as Finite Element Method (FEM) simulations. The study found that body posture mostly depends impact dynamics like head contact timing, projecting how pedestrian behavior in VR can influence real-world vehicle safety design. In road-crossing conditions, movement of person can significantly affect the result of a collision. Meissner et al. [6] and Tang et al. [7] found that fine differences—like walking v/s running—shows how the body reacts on impact. This finding is important for our VR game, where players navigate traffic at different speeds, and accurate body motion helps simulate realistic outcomes. To make the experience even more believable, we also look to Feldstein et al. [8], who showed that using realistic visuals and motion tracking in pedestrian simulations leads to more natural behavior. Their approach on immersion and originality supports our approach of combining true to life animations, interactive environments, and responsive traffic to engage players and reflect real-world behaviour. Real-world parameters like weather and road conditions also play a big role in how safely perdestrains cross streets. Hasirlioglu et al. [9] found that rain and fog can affect how accurately vehicle sensors work, which can influences how both drivers and pedestrians behave. Additionally, Jiang et al. [10] showed that people behave in virtual simulations much like they do in real life—especially when they're interacting with other characters or obstacles. This shows how VR can go over entertainment and be a powerful tool for training and studying real human behavior.

III. Design and System Architecture

A. System Architecture

The core idea behind this simulator is to design and to offer an immersive, customizable and realistic simulation environment by providing dynamic road-crossing scenarios to analyse the pedestrian behaviour. This simulator was developed using Unity game engine which uses real time physics. This simulator is developed with extended comprehensive support for VR Headsets (Meta Quest 3) and comprehensive scenes to recreate traffic situations for analysing pedestrian safety and behavioural studies.

1) Simulator Framework Overview: The simulator is basically structured as a collection of different modules built across the project that connect to replicate real world environments. The environment enables independent testing and development of components such as pedestrian movement, car traffic systems, scenario and level progression, all interacting without any lag within Unity.

Hardware requirements- Windows(64-bit)/ MacOS/ Linux with Intel core i5 or AMD, RAM-8GB or more
Software requirements- Unity installed, C#, XR SDK, Version Control, Visual Studio

B. System Design

1) Environment design: A realistic urban and suburban environment was created using Assets from Unity Asset store. Environment contains a mix of Urban and natural elements. 3D models of objects like research institutes, Financial institutes, retail stores, gas stations, and several other infrastructures were created manually and placed to accurately represent real-world urban layouts. Trees, Plants, Parks and other 3D objects are implemented to experience natural green spaces. key elements such as parking signs, traffic signals, walkways, benches, and streetlights were incorporated to improve the street-level atmosphere. Textures, lighting, and shadows have been incorporated to improve visual authenticity. The sizes of the assets were scaled to ensure that interactions between people and their environment feel accurate. The design includes pedestrian zones, bicycle pathways, and connections, ensuring the simulation environment accurately reflects human behaviour in these areas.

2) Road Design: A realistic road network was built in the virtual world with the help of Road Architect asset. The road design was manually planned and developed in accordance with our design specification and project requirement, providing flexibility in the modelling of the transport system. Roads has been designed using multiple node points.



Fig. 1: Road Design

To achieve environmental realism, pedestrian walkways and cycle lanes were included along with the main roads. Road widths, lane configurations, and surface markings were designed to European Union (EU) road design standards to establish geometric and regulatory realism. Intersections, traffic control signals, and lane guidance systems were also located to manage traffic flow and model realistic traffic patterns. High-resolution road surface textures and realistic colour schemes were applied to support visual realism and provide a more immersive urban setting.

3) Realistic Buildings: OpenStreetMap is used to generate the building layout and shape. We took pictures of each of the building's sides and overlaid them as textures to each of the 3D model's sides.



Fig. 2: Realistic building Created using OSM

We built it one block at a time, matching to the actual building in close detail.

4) Car Simulation: Several car models were taken from MVC Car controller asset and imported. The cars are manually edited using C# and waypoints are attached to cars individually and cars are subjected to follow the waypoints at different speeds given as a variable. Each individual car has a rigid body and a box collider attached in the front to identify a collision. Maximum Speed, Acceleration, Steering Speed, Safe distance between cars can be manually adjusted. A tag 'Car' is assigned to all cars using Inspector menu.

5) Player control and Collision detection: The simulator is based on FPV -First Person View and it uses arrow keys (Forward, Backward, Right, Left) as input to the system. The speed at which the player character is walking can be adjusted variably using a C# script and also enables the simulation of different pedestrian behaviours such as slow walking, running, and fast walking. Mouse cursor input is used for getting data from the screen to look around the screen and to gain situation awareness, which is important for road crossing. The character is attached with a capsule collider and is given tag as 'Player'. When a collider tagged 'Car' hits the collider named 'Player' it counts as a collision. When a collision is detected between a car and a pedestrian, the game is about to end with a fading 'Game over' screen and it replicates an actual fainting after a collision. It means the Player object is unable to control anymore, and the car object will stop immediately and it will take a right turn onto the path and stop.

6) Level Manager: This simulator is designed like an actual game with incorporated levels for increasing the difficulty. The crossing behaviour is approached and recorded in each level and the data can be studied later. There are three levels in this case and Level 1 i.e., 'Easy' has cars approaching slow and the distance between the cars is more. The user has an ample amount of time to cross the road without getting hit. When successfully progressed the first level, User can enter Level 2 which is relatively hard but not so hard. The cars will be approaching fast and the distance between them is less, The user has to walk fast and be very alert to cross the road. Any distractions in this level would lead to collision. The final level is Level 3 and that's the hardest. The cars will be coming very fast and the distance is very less between them. The user has to walk really fast / run to reach the destination.

7) User Interface: A simple UI screen having the Live car speed is shown at the top left corner of the screen. Car Speed will get measured for only the car objects i.e., Objects with the tag 'Car'. Speeds are measured using a Spherical raycast centered on the player and at the radius of 40 units. If any colliders enter the sphere then the object's data like speed, tag is extracted and is fed to the UI. Also, it has a Score shown at the top right corner. Each level starts with a score of 100. Two box colliders with transparent mesh are placed in the road after some distance from the player. Whenever the car hits the collider in the road, 10 points will get deducted and the live score will be updated. The player gets bonus score based on how fast they cross the road.

8) Day and Night cycle: Day/Night cycle and Weather system is an integral part of VR-based pedestrian simulation with the aim to provide a realistic and interactive experience. It generates dynamic weather and simulates the complete day-night cycle. This feature explicitly describes how rain and snow, two common and substantial weather conditions of city settings, would influence pedestrian behavior, visibility, and perceived safety. Rain and Snow can potentially degrade visibility, change the perception of hearing, and lead to additional urgency or confusion on the behalf of pedestrians crossing roads. By adding these conditions, the simulation offers a more enhanced setting in which to evaluate safety, risk-taking, and perceptual accuracy under different weather conditions.

9) Realistic Details: The Birds and Butterfly animations were added to increase the realistic details of the environment of the virtual world. These birds are designed to fly in a flock natural patterns and the direction of the birds flocking and a number of the birds flocking can be customizable across the sky. This gives the developers control over the flow of airborne activity and the density. The Animation of the butterfly also gives a soft, localized motion to the environment which imitates the natural world habitat. Overall these visual elements will give more realistic effects to the game.

IV. GAMIFICATION

A. SCENE MANAGEMENT

To ensure a smooth gameplay experience, the simulator implements a level transition system that advances to the next level automatically whenever player completes the task. This mechanism is custom made using C# scripting and is made to give feedback and completion message once the user successfully completes a level. Each level is built independently using 3D components from asset store. But the components used in each levels are mostly same. Components are usually imported and a realistic environment is being set and is saved to a scene and this scene can be saved as Level 1. The scene is later modified with corresponding changes and saved as another scene named Level 2 and so on Usually each level has a working zone i.e., the region in which the simulator operates mostly. Other environment will be active too but working focus is given to this particular region. In the region, particular zone is set as Level exit zone which has a component with transparent mesh with a collider (here box collider) and with a tag assigned called ‘Level Exit’ tag. Once the player with the tag enters the zone with the tag ‘Level Exit’, the scripting is done to recognize this event as the completion of that particular level. Once the level is completed, next level will get loaded using Unity’s built-in ‘SceneManager.LoadScene’ enabling no manual intervention and to give immersive experience. Also it ensures that the trigger is made active only once per level. Upon successful triggering, ‘Level Complete’ Panel is set to be active. The ‘Level Complete’ panel is just a text with an image in the background and is bound to notify the user that the level has been completed and the next level will be loading. Once a level has been completed, the data will get restored to the scene view again. The ‘Level Complete’ panel also runs in the foreground, notifies the user while loading the next scene in the background.

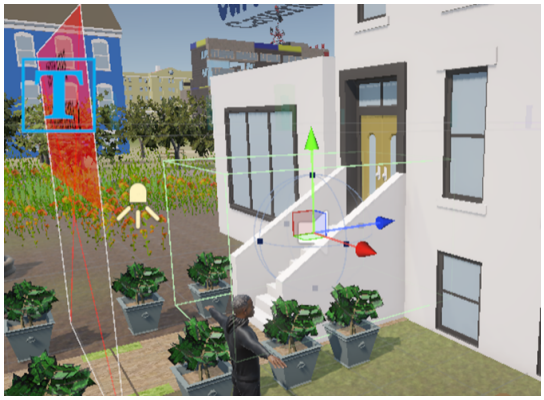


Fig. 3: Box collider with transparent mesh

B. UI Elements

1) Scoring system: When the game gets started, the score is set to be initialized and assigned as 100. The starting time gets stored as input for initializing every time. When game is running, the score card is being updated using a defined method in C# and shown live in the UI panel. When the score card shows the score as ‘Zero’ the game will get over and the user can start the simulation once again. Unlike standard scoring systems, here scores will get deducted for every car the player misses. For every car the player misses, 10 points will get deducted from the score. The system uses a Box collider component with a transparent mesh located after 20 meters from the player on both the ends. When the box collider in the car comes to collide with the transparent mesh collider, a trigger will get initiated and score board will be reduced 10 points every time a car passes through. Only objects with a tag named ‘Car’ will trigger the score card and live score is updated using Update method.

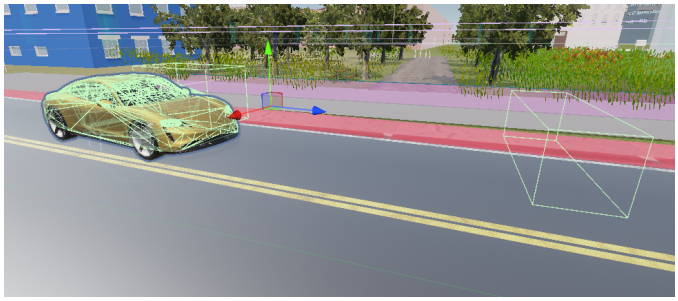


Fig. 4: Transparent mesh colliders for triggering score

2) Live Car Speed Text: Car speed text is kept on the top left corner of the Panel. Car Speed will get measured for only the car objects i.e., Objects with the tag ‘Car’. Speeds

are measured using a Spherical raycast centered on the player and at a radius of 40 units.. Whenever any car object enters the raycast emitted from the player, the car's current speed is extracted from the scripting of the respective car using car.GetCurrentSpeed() function. Unity's standard units (m/s) are converted to km/hr for easy understanding. The conversion can be easily done by multiplying 3.6 with the speed.

3) Arrow Navigation: A 3D arrow mark object is kept at the bottom of the screen, ensuring that it guides the player towards the target. The arrow component is a PNG file basically and it points towards the target while navigating. The scripting has been done to get the reference from the Player and the target. In each frame, the normalized direction vector is computed towards the target. Angle has been calculate every frame and rotation will be done in the z direction to make the arrow rotates clockwise or counterclockwise, pointing toward the destination.

4) 'Game Over' Animation: While crossing the road, if the player gets hit by a car at any instance the simulation will be ended with a animation showing the game is over. The scripting has been done to figure out the colliding component's tag. If the tag is set to be 'Car' then it is a collision that triggers the sequence. Once collided, the player control will be turned off and character and camera rotations are done using 'Quaternion.Lerp' to simulate movements. After a short delay, a transparent image will darken the screen and a text 'GAME OVER' will appear to notify the player.

5) Pedestrian animation: For creating a lively environment, non-player characters are used to perform several activities like walking, talking, using ATM, listening music, sitting in the park, etc. This method uses NPC models and animations from Mixamo, which are chosen for Unity compatibility. After importing these assets into the Unity, these animations are configured with a humanoid rig, making human-like movement and visualization a reality. Non-player characters entering into the buildings, staying inside for a bit, and returning to interact with others adds to the natural flow. C# scripts handle character entry and exit logic for buildings, enabling timed navigation and transitions. Conversation scenes use delays and sequence management in Unity, so when one character talks, the other responds after a controlled delay, creating a natural exchange. The text displays with controlled delays makes UI texts appear gradually, improving readability and natural flow. All these together created an realistic experience that is both sensible and engaging.

C. NPC Spawner

In Unity, a NON PLAYER CHARACTER(NPC) system was created to simulate the pedestrian in virtual environment.To enable pedestrian movement, multiple planes were created to define the paths and walkable areas.Each of these planes were assigned to a custom Unity layer labeled "Footpath" for NavMesh baking purposes.

A NavMesh modifier volume component was added to each plane,to ensure Only designated planes are walkable excluding all other areas in the environment.To manage Navmesh generation a separate Empty Game Object was created and assigned with a Navmesh Surface component.Upon baking the planes turn light pink(sometimes blue) as shown in Fig.5 indicating they were baked with Navmesh.

After baking planes, we need characters,For this purpose, Characters and their Animations were sourced from Mixamo.com and assigned each character prefab with a Box collider for collision detection, a rigid body component, an Animator component to control the animation of the characters,a Navmesh agent for pathfinding and a controller script to control the movement of pedestrians.

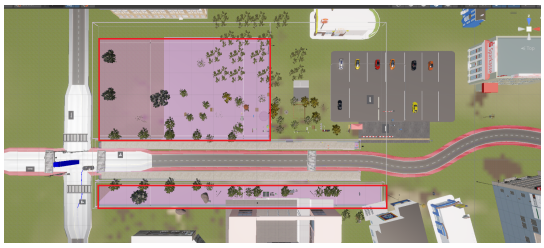


Fig. 5: Baked planes

A spawner script has been assigned to the main camera.All the character prefabs were grouped and the script assigns destinations and randomly choose spawn points and if a character reaches destination the script assigns new destination.With this arrangement,the NPC's can independently spawn in a specified area.

D. Pedestrian sounds

In Unity-based game development, sound is a quiet yet powerful storyteller that pulls players deeper into the world and helps them feel they really belong there. This becoming even truer in augmented and virtual reality, where ears and eyes work together to sketch the surroundings players explore.

We walked into the audio work with a simple mission: make every scene believable by blending background with sounds that follow moving objects. To ground the player, we added footsteps, chattering crowds, and the sounds of bus and car engines then dotted them around the space so they echoed the players location and actions.

Sound plays a large role in the determination of how an environment is perceived, particularly in VR. Voice and sound can be represented realistically by Unity, with sound effectively responding to the player's movements.The player's footsteps might provide the player with an enhanced sense of presence and movement, and wind or engine noise might help determine the direction and distance of a car. Sounds happening in the distance, such as a conversation or crowd

noise, add character and atmosphere to the environment, instead of it being silent or artificial.

This approach creates a more authentic city-like atmosphere. As a result, the sound design helped immerse the player in the scene, making the game feel more realistic and engaging.

E. AUDIO EFFECTS

Previous studies showing that sound can affect how we perceive visual scenes—and recognizing the general underrepresentation of auditory elements in virtual environments and games [11]. To enhance realism in pedestrian simulations, audio is crucial for compensating for visual constraints. By using 3D audio, users gain better situational awareness, allowing them to detect elements beyond their line of sight, which in turn boosts immersion and supports more realistic behaviour [12].According to [13] audio has a significant impact on how we interpret visual information, yet it remains an underutilized element in many virtual environments.As video games continue to grow in popularity, there has been an increasing trend toward using game-based technologies for educational and training purposes, often referred to as serious games [13].

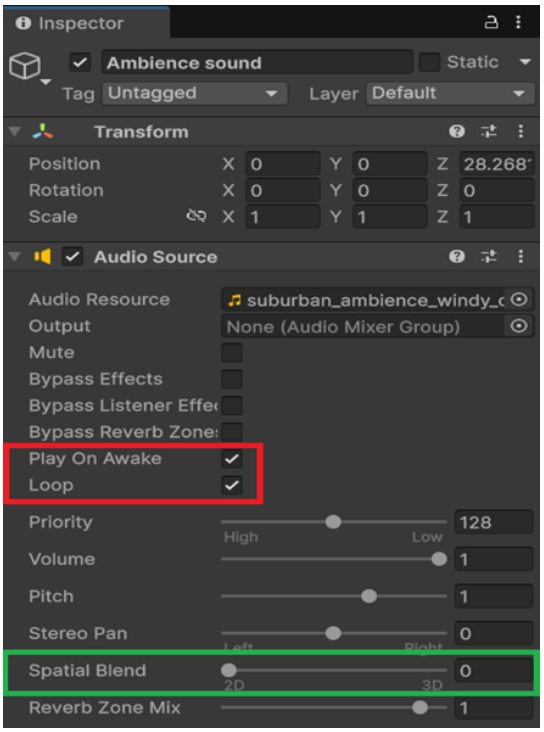


Fig. 6: Unity settings for ambient sound with loop, play on awake, and 2D spatial blend for background audio.

Auditory Stimuli The auditory stimuli in our project consist of three distinct conditions: (1) Environmental ambient sound including distant traffic and wind, (2) Moving Cars, Engine Audio Effects, and (3) Idling Bus Sound with Distance-Based Attenuation. Their audio files are downloaded for free from Pixabay.com A. Environmental ambient sound To enhance the natural immersion of the outdoor simulation environment, continuous ambient sound was implemented to represent a typical street-side setting. This included looping audio that combined air breeze, birds chirping, and tree leaves rustling—providing a calm, realistic backdrop and helping prevent sensory isolation.

In Unity, this was achieved by attaching an Audio Source component to an empty GameObject labelled "Ambience sound.",The Loop and Play On Awake checkboxes (shown in the red box) were enabled so the sound starts automatically and plays continuously during the simulation. The Spatial Blend slider (highlighted in green) was set all the way to 2D, ensuring the ambient sound remains consistent and non-directional, no matter where the player moves or which way they're facing.

Moving Cars, Engine Audio Effects To create a lively simulation of multiple moving vehicles, every car was equipped with a 3D spatial engine sound. This makes the pitch and volume of the engine change depending on their speed and distance to the game player, effectively mimicking the Doppler effect. This was accomplished in Unity by attaching an Audio Source component to

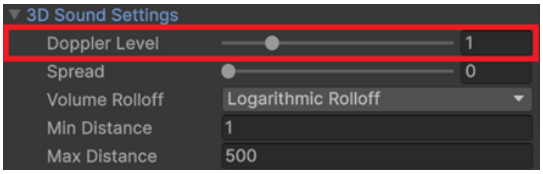


Fig. 7: Doppler Level set to 1 for realistic pitch shifts in moving audio sources

every vehicle and enabling 3D sound settings through the Inspector.Ensuring fully 3D spatial audio,which ensures that the volume of cars changes with respect to change in distance from game player. C# script (movingcarsound) is assigned to manage real-time audio behaviours, depending on cars speed and proximity. Doppler Level is set to 1, which enables the natural shift in pitch as vehicles move towards or away from the game player. This makes sure to replicate the way a real engine sounds changes due to the Doppler effect, which adds life to the simulation. These configured audio components create realistic soundscape, which allows players to hear and interpret vehicles movements without visual contact. Thus, it enhances situational awareness and realism.

1) Idling Bus Sound with Distance-Based Attention: A stationary bus was placed at a virtual bus stop to simulate realistic engine idling noise, enhancing the street scene's authenticity.An Audio Source component was attached to the bus Game Object. The sound is set to Loop, ensuring continuous playback. Spatial Blend is set to 1, making it fully 3D so the audio reacts to the player's position. The Doppler Level is also set to 1 to simulate natural pitch shifts if there were movement. The Volume Rolloff is configured as Logarithmic, allowing a natural fade over distance. The sound becomes inaudible beyond 50-meter range, as shown in the volume curve.This creates an immersive sound zone that increases in volume as the game player approaches, which contributes to the sense of realism in the environment.

V. Meta Integration

To enable immersive VR experience,the Environment was developed using Unity game engine and Meta's XR softeware development kits(SDK's).The Oculus XR was configured through Unity's XR plugin management system for Android,Which enables Runtime support for quest-3 as Meta Quest-3 Runs on Android platform.Three primary SDKs were integrated: the Meta XR Core SDK ,Which supports the Device for Rendering and the Meta XR Interaction SDK to take input from hands and controllers, and the Meta XR Movement SDK for enabling full-body tracking.These packages were directly imported to the project from the Unity's Package Manager.This foundation ensured the project could support the immersive rendering, interaction, and motion tracking. One of the fundamental requirements for the simulation was natural user embodiment with full-body tracking. This was provided with the help of the Meta XR Movement SDK utilizing AI-powered pose estimation and Quest onboard sensors to facilitate skeletal tracking. Initial steps involved the import of a human-like character model with the help of Unity's Avatar system and verifying the same for bone buildup and rig compliance. Degrees of Freedom (DoF) translation was activated to enable positional motion within the virtual space. With the Retargeting Config Editor from within the SDK, T-pose calibration occurred and the skeleton of the avatar aligned with Meta's motion-tracking system. The OVR Camera Rig prefab was affixed to the character hierarchy with its transform reset to move it into correct positional alignment with the user's actual head direction and body orientation.Whole-body tracking assets OVR Body and OVR Skeleton Retargeter and OVR Unity Humanoid Retargeter were utilized with the avatar. "Full Body" tracking mode was chosen to ensure that head, torso, arm, and leg motion were accurately represented within the simulation in real time.

VI. Experimental analysis

A series of experiments have been carried in Unity with varied Pedestrian walking speeds and the same being applied across all the levels. The simulation carried out may vary if the user is employed with Meta Quest and the data will vary further according to each and every volunteer who is willing to take part in testing out. Volunteers of various age groups will be taken into consideration and certain distractions will be applied to each and so. For our experiments, The width of the road is set to standard 8 meters. We have changed the walking speed of the Player for test in each level. Initially, the speed was set to 1m/s (slow walking) and the initial Car speed was set to 8m/s and the simulation has been carried out. It turned out that the Player took 8 seconds to cross the road. In the first case, Cars are set to approach only in a single lane. Player is assumed to cross the road comfortably without colliding. Based on the data from the simulation, further calculations are carried out using the formulas.

$$\text{Car Arrival Time} = \frac{\text{Distance between car and the player}}{\text{Speed of the car}} \tag{1}$$

$$\text{Time required to cross} = \frac{\text{Road width}}{\text{Walking speed}} \tag{2}$$

$$\text{Margin of Safety (s)} = \text{Car arrival time} - \text{Time required to cross} \tag{3}$$

Margin of Safety is basically to find out the possibility of crossing the road successfully without any collision and it comes with a value. Higher the value of the margin the safer the player crosses the road. In this table, We have calculated the margin tentatively based on time required and any further distractions in the case can be tested out and analyzed later. Also, While allowing the cars to run in both the lanes will lead to decreasing distance between the cars and the margin of safety will be reduced accordingly.

TABLE I: Margin of Safety (in seconds) at Varying Car Speeds and Walking Speeds

Car Speed (m/s)	1 m/s	1.4 m/s	2.6 m/s	3.5 m/s	4.8 m/s
8	2.00	4.29	6.92	7.71	8.33
10	0.00	2.29	4.92	5.71	6.33
12	-1.33	0.96	3.59	4.38	5.00
15	-2.67	-0.38	2.25	3.04	3.66
22	-4.36	-2.07	0.55	1.34	1.97
28	-5.14	-2.85	-0.22	0.59	1.19

In higher levels, the distance between the car and the player will be reduced and speeds will be increased. Hence the margin will be varied very much. Unless there is a change in the walking behaviour of the player it will be really difficult to reach the destination.

VII. Future Scope

This virtual reality game has the potential to significantly improve traffic safety, particularly for pedestrians. It might teach children in schools how to cross the street safely in an enjoyable and secure manner. There isn't any actual risk when practicing because everything takes place in virtual reality. Additionally, it could boost the confidence of elderly or injured individuals before they return to the outdoors.

The game's ability to depict how people move in various traffic scenarios may potentially be useful to city planners in the future. It might be used by researchers to observe how individuals respond to distractions, fast autos, and self-driving cars. The game's user base may provide valuable information for bettering traffic signals and street layouts. This VR tool has the potential to be utilized for city planning, safety training, and perhaps treatment with further development, demonstrating how a game can improve the real world.

VIII. Conclusion

The suggested VR pedestrian simulator is useful for developing a real-world setting in which users can be exposed to actual street-crossing scenarios by interacting with dynamically moving traffic.Interactive aspects of the simulator, including a live scoring system, vehicle speedometer, and collision animation, provide feedback in real time, increasing user engagement and supporting learning. Integrated environmental audio effects, including car engines, city ambiance, and dynamic sound attenuation based on player distance, add realism to the simulation, a critical requirement for user immersion in VR environments. The system is also set to measure pedestrian behavior, capturing crossing time information, collision incidence, and user preference for different scenarios. The data can be useful for decision pattern making and inform safety design in urban planning and autonomous navigation systems. Additionally, the simulator is also usable as a training tool to raise the level of awareness of pedestrian safety, particularly among children and vulnerable road users, via simulations in a risk-free setting for safe crossing practices. The modularity approach, realized by Unity and Meta Quest integration, is supportive of future extension, e.g., via the addition of weather conditions, distraction situations, or AI-driven pedestrian models for more sophisticated interaction.

Overall, the VR pedestrian simulator is an effective integration of system engineering and game design methodologies to develop solutions to real-world traffic safety issues and to provide a research arena to study pedestrian behavior in highly controlled, immersive settings.

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